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Vision-Based Apple Counting and Yield Estimation

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Abstract. We present a novel method for yield estimation in apple orchards. Our method takes segmented and registered images of apple clusters as input. It outputs number and location of individual apples in each cluster. Our primary technical contributions are a representation based on a mixture of Gaussians, and a novel selection criterion to choose the number of components in the mixture. The method is experimentally verified on four different datasets using images acquired by a vision platform mounted on an aerial robot, a ground vehicle and a hand-held device. The accuracy of the counting algorithm itself is 91%. It achieves 81–85% accuracy coupled with segmentation and registration which is significantly higher than existing image based methods.

1 Introduction

For specialty farms such as apple orchards, crucial decisions such as choosing the number of pickers, the number of storage bins and early season sales contracts depend on estimating fruit count. In this paper, we present a method for counting apples from images of an orchard row. The images can be captured from various types of robotic and non-robotic platforms such as ground or aerial robots or a person carrying a hand-held camera.

Counting apples from a running sequence of images is difficult for three main reasons. First, there is a trade-off between image footprint and the size of individual apples in the image. If we use close-up views, coverage becomes tedious. On the other hand, if we use a wide-angle camera, individual apples occupy only a small area in images. Second, apples can be found in arbitrarily shaped clusters in which almost all apples overlap with each other. Segmenting individual apples from such cluster images is challenging. Third, because of occlusions due to leaves and branches as well as specularities sometimes apples are not detectable at all.

Surprisingly, most robotic yield estimation systems do not focus on accurate counting. Instead, these systems rely on developing a consistent estimator of actual fruit count. These methods include Wang et al. [1] who used morphological operations and ellipse fitting for counting. Their method does not handle more than two apples in a cluster. Hung et al. [2] relied on circular Hough transformation to extract individual apples. Linker et al. [3] presented a method

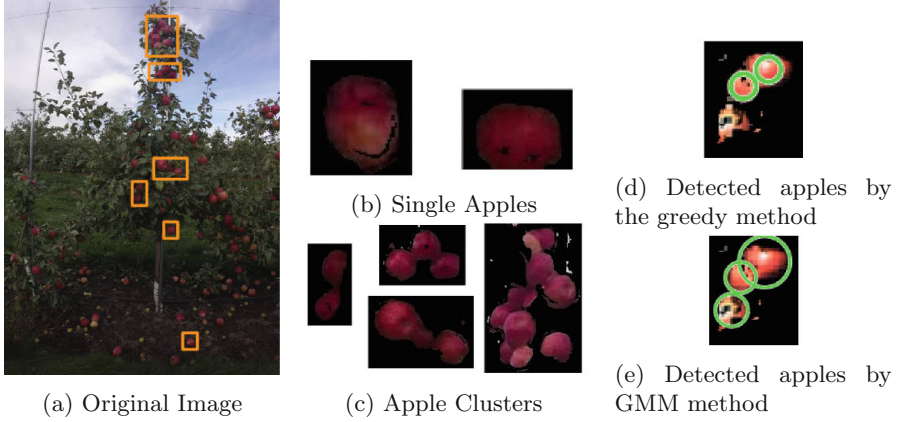


Fig. 1. A sample image and few example apple clusters. Figure (b) shows two example single apples. These apples can be detected easily by state of the art methods such as circular Hough transform or ellipse fitting. It is hard to generalize these methods for more complex clusters shown in figure (c). Figure (d) shows a sample output from the greedy circle fitting method. Figure (e) shows the output of GMM method on the same input.

based on fruit edge detection. This method requires very high resolution images. All these techniques work for simple instances such as those shown in Fig. 1b. In more challenging, yet common instances (see Fig. 1c), both Hough transform and ellipse fitting will fail. To identify the number of apples correctly and to locate individual apples from these images we need novel models and techniques. In this paper, we present two such methods: one for counting the apples based on a classic clustering technique - Gaussian Mixture Models (GMM) [4] and another using an intuitive greedy circle fitting algorithm. The clustering technique is robust and accurate. The greedy method is identical to the state of the art techniques such as circular Hough transform and ellipse fitting. We use it as a baseline for comparison with the performance of the GMM method.

Our method takes as input segmented apple images where non-apple pixels have been eliminated. We also assume that the images have been registered and apple clusters have been aligned across images. The details of segmentation and registration of apples across multiple images can be found in our earlier work [5,6]. In this paper, we focus on the following problem: Given a set of segmented registered apple images, count the number of individual fruits in the images and identify their locations. We start with our technical approach in the next section.

2 Technical Approach

Given a set of segmented and registered input images, we want to find the number and location of all the apples in every image. In this section, we propose

two different methods for solving this problem: first using Gaussian Mixture Models (GMM) and second based on greedy circle fitting. The greedy circle fitting approach provides a baseline for comparison. As the first step for both methods, we perform a connected component analysis on the input image and detect all the individual clusters. We compute the location and radius of each apple in each cluster. As the clusters are disjoint the total number of apples can be obtained by simply summing up the numbers across all clusters.

2.1 Gaussian Mixture Model Method

In the GMM method each apple is modelled by a Gaussian probability distribution function (pdf) and apple clusters are modelled as mixture of Gaussians. We start by converting the input cluster image I to binary. Let this binary image be denoted by I_b . The locations of the non-zero pixels in the binary image are used as input to GMM.

Let X represent the set of apples we are trying to find. Then, we can convert our problem to a Gaussian mixture model formulation in the following way:

$$P(I_b|X) = G^k(\phi, \mu, \Sigma) = \sum_{i=1}^k \phi_i G_i(\mu_i, \Sigma_i) \quad (1)$$

Here, $G^k(\phi, \mu, \Sigma)$ is a Gaussian mixture model with k components, and G_i is the i th component of the mixture. μ_i and Σ_i are the mean and covariance of the i^{th} component. The covariance matrix $\Sigma_i = [\sigma_{x_i}^2, \sigma_{y_i}^2]$ is diagonal. ϕ_i is the weight of the i^{th} component where $\sum_{i=1}^k \phi_i = 1$ and $0 \leq \phi_i \leq 1$.

Given model parameters $\theta = \{\phi, \mu, \Sigma\}$, the problem of finding location of the center of the apples and their pixel diameters can be formulated as computing the world model which maximizes $P(I_b|X)$. Each component $G_i(\mu_i, \Sigma_i)$ of the mixture model represents an apple with center at μ_i , equatorial radius $2\sigma_{x_i}$ and axial radius $2\sigma_{y_i}$.

A common technique to solve for $\arg \max P(I_b|X)$ is the expectation maximization (EM) algorithm [4]. As is well-known, EM provides us a local greedy solution for the problem. Since EM is susceptible to local maxima, initialization is very important. We used MATLAB's [7] implementation of K-means [8] (which uses randomly-selected seeds to avoid local maxima) for initialization of EM.

Selecting the Number of Components: In our problem formulation, the number of components k is the total number of apples in image I . EM enables us to find the optimal location of the apples given the total number of apples k . Our main technical contribution in this paper is a method to calculate the correct k . Let the correct number of apples in the input image be denoted by κ . We tried different state of the art techniques (Akaike Information Criterion (AIC) [9], Minimum Description Length (MDL) [9] etc.) for finding κ . None of them worked out of the box for our purposes (Fig. 2). Therefore, we propose a new heuristic approach for evaluating mixture models with different number of components based on MDL. Unlike classic MDL based approaches we have both reward and penalty.

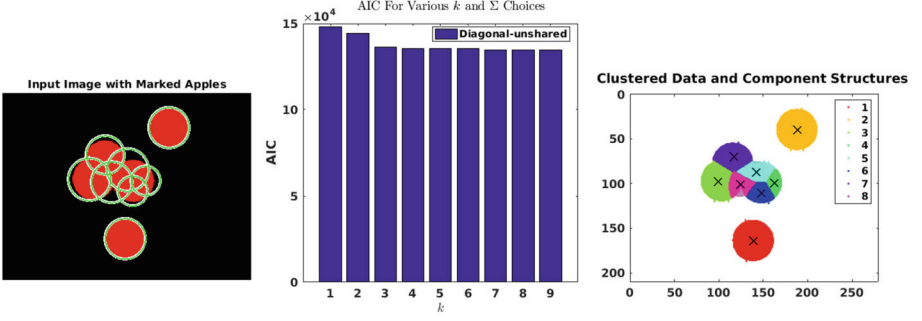


Fig. 2. Popular methods like AIC/BIC do not work out of the box for our purposes. These criteria have a tendency of choosing higher values of k . In this synthetic image, we have only five circles but the AIC based criterion value (Bar plot in the middle) was lowest for $k = 6$ and consequently it finds eight apples.

Let $\sigma_{min} = \min(\sigma_{x_i}, \sigma_{y_i})$ and $\sigma_{max} = \max(\sigma_{x_i}, \sigma_{y_i})$. Using the mean and covariances of the i th component we define a 2D Gaussian kernel $\mathcal{G}(\mu_i, \sigma_{max})$ where σ_{max} is the variance. Let $P(\mu_i)$ denote the response of the kernel when placed at the center μ_i in the original input image I and C_i denote the total number of pixels clustered by $G_i(\mu_i, \Sigma_i)$. For each component $G_i(\mu_i, \Sigma_i)$, of the mixture model $G^k(\phi, \mu, \Sigma)$ we define the reward R_i in the following way,

$$R_i(G_i) = \phi_i \left[P(\mu_i) + P(\mu_i) \left(\frac{\sigma_{min}}{\sigma_{max}} \right)^2 + P(\mu_i) \frac{C_i}{\pi \sigma_{max} \sigma_{min}} - \frac{1}{3} (\pi \sigma_{x_i} \sigma_{y_i} - C_i) \right] \quad (2)$$

For most of the images we only capture the frontal views of the apples, which can be easily approximated by circles lying on a plane. All four terms in Eq. (2) reward specific spatial characteristics of the Gaussian pdf related to this fact. $P(\mu_i)$ represents the strength of the distribution in terms of pixel values and is present in first three terms. The second term rewards circular shaped distributions using the eccentricity of the pdf. As the eccentricity $\epsilon = \sqrt{1 - \frac{\sigma_{min}^2}{\sigma_{max}^2}}$, and for circles ϵ is zero, we use $1 - \epsilon^2 = \left(\frac{\sigma_{min}}{\sigma_{max}} \right)^2$ as the rewarding factor. The third term rewards coverage. The fourth term penalizes Gaussian pdfs covering large area and clustering very few points.

Now if we find out the reward $R_i(G_i(\mu_i, \Sigma_i))$ for all the components k , the total reward for the mixture model $G^k(\phi, \mu, \Sigma)$ can be computed by summing them together.

Next, we define the penalty term. The traditional MDL penalty term is $U = cp \log(|Y|)$ where p is the number of parameters in the model, $|Y|$ is the total size of the input data, and $c = \frac{1}{2}$ is a constant. Based on this principle, our penalty term is $V(G^k(\phi, \mu, \Sigma))$ is defined as the following

$$V(G^k(\phi, \mu, \Sigma)) = c'(3k) \log\left(\sum_{\mathbf{x}} (I_b(\mathbf{x}) \neq 0)\right) \quad (3)$$

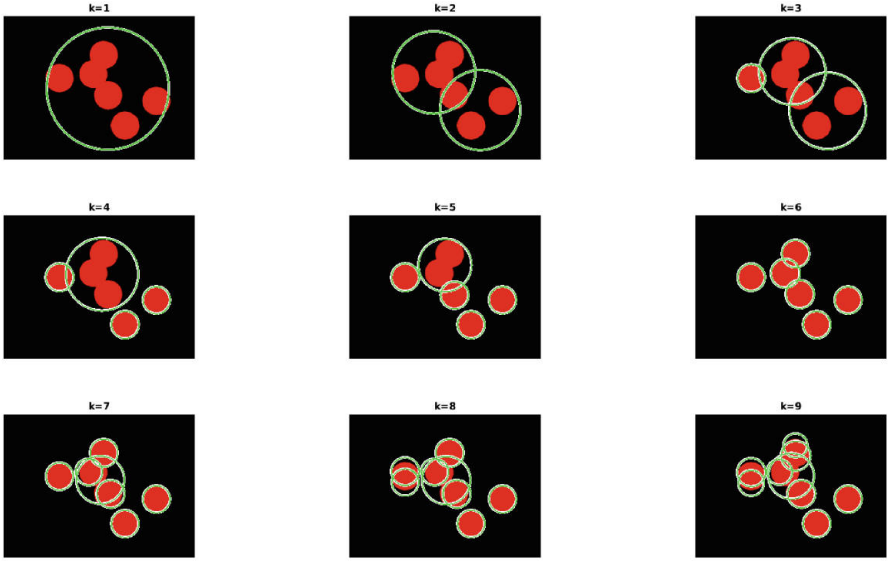
where $\mathbf{x}$ represents the pixel index across the image I_b . Compared to traditional MDL based penalty we have the constant $c' = \frac{3}{2}$ instead of $c = \frac{1}{2}$. This is attributed to the fact that the reward expression (2) has three terms compared to one. The number of components k is multiplied by three as each Gaussian has three parameters $[\mu_i, \sigma_{x_i}, \sigma_{y_i}]$. With these terms defined, we choose the correct number of components κ in the following way:

$$\kappa = \arg \max_k R(G^k(\phi, \mu, \Sigma)) - V(G^k(\phi, \mu, \Sigma)) \quad (4)$$

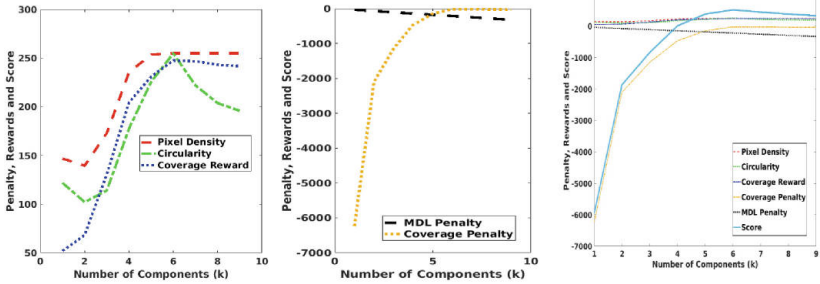
To have a better understanding of the selection procedure, we demonstrate a synthetic example at Fig. 3(a). From Fig. 3(b), it is evident that except for $k = 6$, other mixtures have low circularity. The coverage rewards and pixel density components increase with k and converges to a steady state. While the penalty from minimum description length principle increases with k , generally the penalty for coverage decrease with k . In this example, the crucial factors in determining the score are circularity and the coverage penalty. For $k = 6$ circularity is at the peak and coverage penalty is lowest and consequently the score was maximum for $k = 6$. The plot of the corresponding rewards, penalties and final scores are shown in Fig. 3(b)–(d). We show sample results from our datasets in Fig. 4.

2.2 Greedy Circle Fitting Method

The GMM method uses a global fitting criterion to segment the apples. In this section, we present a greedy approach which achieves reasonable performance efficiently. In case of images containing single apple, with rough knowledge of apple radius in terms of pixels, we can create Gaussian kernels of different sizes within the known bounds, convolve the entire image with them and find the maximum response location to find the center of an apple. This concept can be extended in case of images containing multiple apples. The first apple is chosen as the apple with maximum kernel response. Likewise, the second one can be computed by choosing another apple, which combined with the previously chosen one maximizes the total response of both. In this method, the order in which the apples are selected defines an occlusion hierarchy. The response of the combination multiple Gaussian kernels are computed using this hierarchy. Figure 1(d) shows a detected apples on an input image from the greedy method. It is unable to detect the partially visible apple. Figure 1(e) shows the output of GMM method on the same input. It successfully finds the apple missed by greedy method.



(a) Predicted circles for different number of components in GMM



(b) Rewards for different number of components (c) Penalties for different number of components (d) Final score for different number of components

Fig. 3. A synthetic example consisting of six random circles and plots illustrating how the number of components are selected. Figure (a) shows how the pdfs cover the circles for different values of k . Figure (b)–(d) shows how the score calculated from the rewards and penalties following the right hand side of Eq. (4). The plot shows that the score is maximum for $k = 6$ which is indeed the correct number of components.

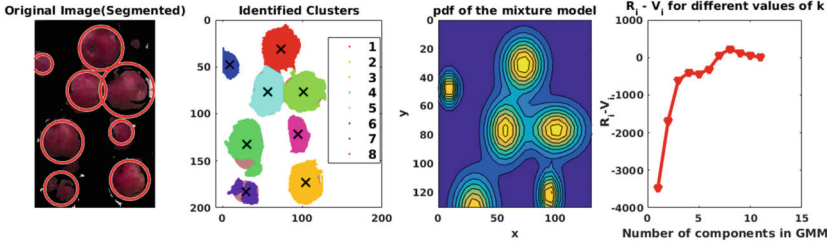


Fig. 4. A sample output from GMM on real data.

3 Experimental Results

We have evaluated the performance of our counting algorithm on four different datasets. These datasets are obtained from different platforms. The first two datasets are obtained from hand-held cameras, the third dataset from a camera mounted on a ground vehicle and the fourth dataset from a camera mounted on a flying UAV. Figure 5 shows sample images from all four datasets.



Fig. 5. Sample images from used datasets. The two images on the left are obtained from hand-held cameras. The third image from left is obtained from a camera mounted on a ground vehicle and the rightmost image is obtained from camera mounted on a UAV.

We first evaluate the GMM method with the segmented apples clusters from Dataset 3 (Fig. 1(b), (c)). We used 442 such images for this purpose, all of which were hand counted. We identified clusters of different sizes (Fig. 6(a)) and evaluated the performance of our algorithm in terms of accuracy, over counting and under counting per cluster size (Fig. 6(b)). The over all accuracy for the entire dataset is 91.30%. For clusters containing six or more apples, the percentage of under counting goes up to 33.33%. The number of instances of such large sized clusters were very low to make any strong inference about the performance of the algorithm. We need further investigation to analyze the performance of the algorithm in these cases. For single apples, we found 10% over counting.

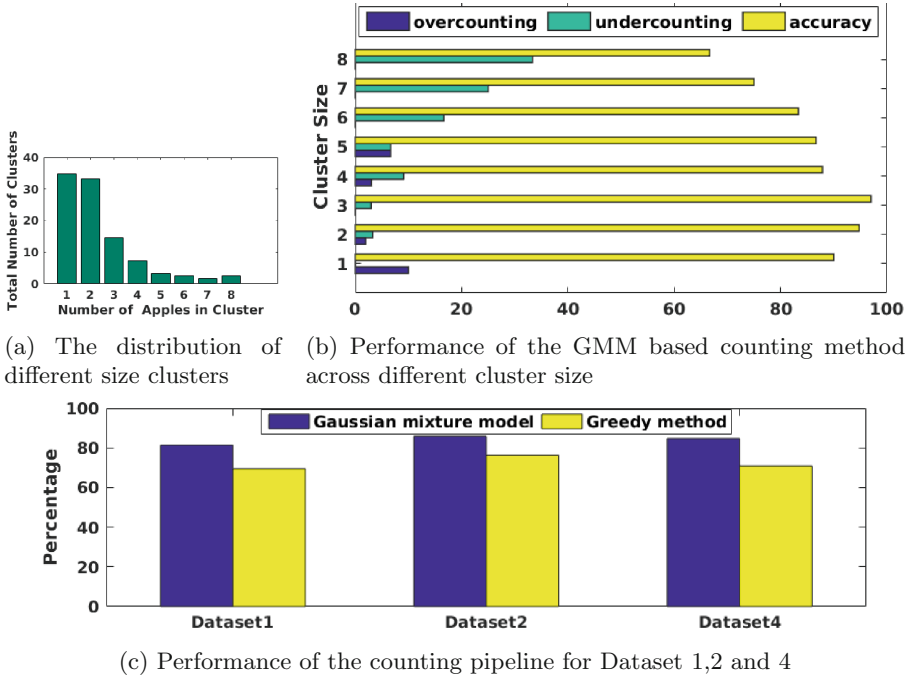


Fig. 6. Performance evaluation.

This problem can be attributed to occlusion and specularities which make the algorithm think that there are two apples instead of one.

Next, we evaluate the gross performance of our counting pipeline using the Dataset 1 2 and 4. The gross results for Dataset 3 were not available at the time of submission. Dataset 1 contains a mixture of red and green apples. The total number of images in this dataset is 464. It covers a block in an orchard row containing six trees. Dataset 2 primarily contains red apples. The total number of images in this dataset is 964. It covers a full row in an orchard. Dataset 4 was collected using an UAV. It contains 655 images and covers six trees. The total number of hand counted apples from the images in the datasets are 258 and 952, and 673 respectively. The accuracy of GMM counting method for first and second and fourth dataset is 81.3492%, 85.87% and 84.72%. The drop in accuracy for the first dataset is mainly attributed to our segmentation method [5] where many green apples were not detected. Figure 6(c) shows a bar chart of the accuracy of our counting methods. We show the accuracy from the greedy method for comparison. For the greedy counting method the accuracy drops down to 69.44% for the first dataset and 76.34% for the second dataset. This drop is expected as the greedy method uses a hard threshold for grouping apple pixels. It fails when the scale of the image varies a lot or apples are only partially visible.

4 Experimental Insights and Conclusion

In this paper, we presented a method for counting apples from segmented and registered input images. While the results are encouraging, there is still room for improvement. Specifically, since the results presented here constitute the final part of a three step pipeline, it is important for the previous steps to be correct. For example, in Fig. 7(a), the counting procedure did not count any of the green apples. The reason behind this is that they were not detected by our color based segmentation method. The algorithm also fails to predict the correct number of fruits when there is too much overlap and boundaries among different fruits are not detectable. To get better insight into this problem, we created a synthetic input where the algorithm fails (Fig. 8). Here, the predicted number of components by the algorithm is five but the actual number is six. A closer look at the solution from EM algorithm for $k = 6$ shows that, actually EM failed to find out the correct location and pixel radius of fruits which resulted in large coverage penalty and low circularity. These types of problems are present in natural conditions too. One such observation is shown on Fig. 7(b). For these types of scenarios, it is hard to predict the correct number of apples from a single view. A natural way to resolve these ambiguities is to use active vision (look at the cluster from another viewpoint). In future, we would like to use the information obtained from single views, to move actively in more suitable positions and fuse all the previously obtained information to predict the correct number and locations of fruits.

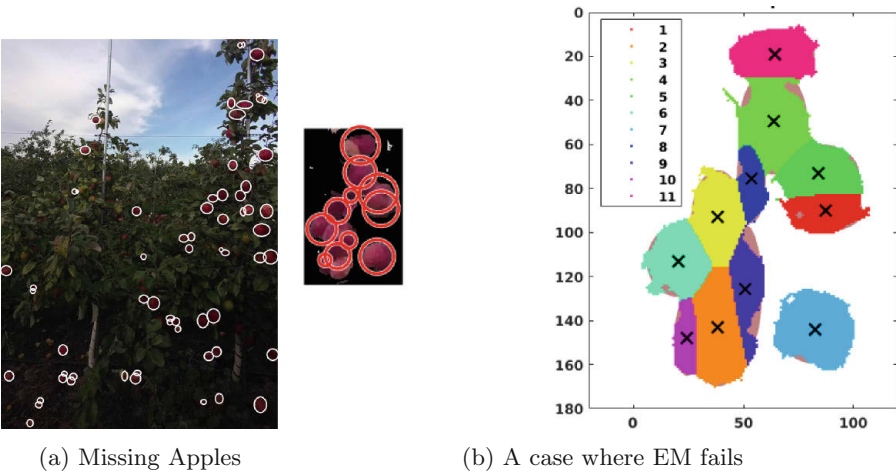


Fig. 7. A few failure cases in real data.

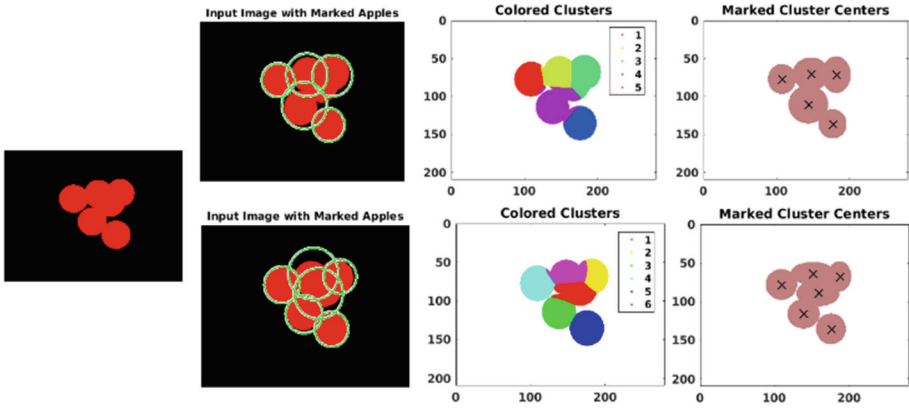


Fig. 8. A synthetic input where the algorithm fails to compute k correctly. If we look closely at the solution found by EM for the correct k , we find that EM failed to find the correct location which resulted in large coverage penalty and low circularity.

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